

# Vladimir Strokov

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## EDUCATION

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**Johns Hopkins University** Baltimore, USA  
Department of Physics and Astronomy  
**Ph.D.**  
Advisor: Emanuele Berti / Co-advisor: Giacomo Fragione (Northwestern) 2020–2024

- Research: Gravitational Waves, Intermediate-Mass Black Holes, Populations of GW Sources

**Lebedev Physical Institute** Moscow, Russia  
Astro Space Centre, Theoretical Astrophysics Department  
**Candidate of Sciences**, Advisor: V.N. Lukash 2011

- Research: Gravity Theory

**Moscow Institute of Physics and Technology** Moscow, Russia  
Ginzburg’s Chair for Problems of Physics and Astrophysics  
**M.Sc.**, *with excellence* / **B.Sc.**, *with excellence* 2007 / 2005

## RESEARCH

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**West Virginia University** Morgantown, USA  
Center for Gravitational Waves and Cosmology (GWAC)  
**GWAC Postdoctoral Fellow**, LISA GW science 2024–present  
Activities:

- LISA data analysis: global fit problem, denoising techniques
- Double white dwarfs in LISA: confusion foreground, source mixing; coordinator of two proposed LISA Consortium projects
- Black hole shadows: image recovery, machine-learning methods

**Lebedev Physical Institute** Moscow, Russia  
Astro Space Centre, Theoretical Astrophysics Department  
**Lab Assistant / Research Fellow** (on medical leave: 2014–2017) 2005–2020  
Activities:

- GPU algorithms for simulations of black hole shadows
- Extraction of black hole silhouettes from future interferometric data
- Selecting exoplanetary candidates for the Millimetron project

**Federal University of Esp rito Santo** Vit ria, ES, Brazil  
J.C. Fabris’ Gravitation and Cosmology Group  
**Postdoctorate**, theories of modified gravity 2013–2014  
**Federal University of Juiz de Fora** Juiz de Fora, MG, Brazil  
I.L. Shapiro’s Field Theory Group  
**Postdoctorate**, field theory in curved space–time 2012–2013

## PROFESSIONAL MEMBERSHIPS

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- LISA Consortium (member since 2020; **core member** since 2025)
- American Physical Society (since 2021)
- American Astronomical Society (since 2024)
- International Society on General Relativity and Gravitation (since 2022)

## TEACHING

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- Sole Course Instructor for Advanced Electromagnetism 1 (Spring 2026) at *West Virginia University*
- Teaching assistant at *Johns Hopkins University*: graduate-level Electrodynamics (Spring 2022–2024), graduate-level Quantum Mechanics I (Fall 2022), General Physics Lab II (Spring 2021), General Physics for Physics Science Majors (Spring 2021)
- Developing and teaching an intersession course at *Johns Hopkins University*: “MCMC: a lever to move the world of data” (Intersession 2023)
- Developing and teaching physics “bootcamps” for the Post-Baccalaureate Premedical Program at *Johns Hopkins University*
- Physics tutor for the Post-Baccalaureate Premedical Program at *Johns Hopkins University* (2022–2023 academic year)
- Mentoring a high school student (Jacob Winick, Millburn High School, NJ) on a research project *Visualizing Gravity in GR* (2022)
- Undergraduate-level course “Introduction to Cosmology” (2017–2019), *Moscow Institute of Physics and Technology*
- Undergraduate-level courses “Introduction to Astronomy” (Spring 2013) and “Introduction to General Relativity” (Fall 2013), *Federal University of Esp rito Santo*
- Minicourse “The Standard Cosmological Model and Its Observational Basis” (Spring 2012) and the core program Mathematical Physics II (Fall 2012), *Federal University of Ju z de Fora*

## AWARDS AND GRANTS

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- ACCESS Explore allocation for the project “Reconstructing black hole shadows from sparse interferometric data” (200,000 ACCESS credits  $\approx$  200,000 core-hours)
- Space@Hopkins Seed Grant “Preparing to Untangle the Gravitational Wave Cacophony from Multivariate Data for the LISA Space Mission” (PI: E. Berti, Co-I: S. Kushnarev, Collaborators: **V. Strokov**, C. K ummerle, N. Geissler; \$25,000)
- 2023 Teaching Award (Department of Physics and Astronomy, JHU)

## PROFESSIONAL SERVICE

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- Referee for *Physical Review D*, *Classical and Quantum Gravity*, *European Physical Journal C*, and *Astronomy & Astrophysics* (20 manuscripts refereed, 2017–present)
- Session chair: APS April Meeting 2024, Sacramento, CA; NANOGrav Fall Meeting 2025, Missoula, MT
- Local organizing committee, II Jos  Pl nio Baptista School of Cosmology, Pedra Azul, Brazil (2014)

## PROGRAMMING PROFICIENCY

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The scientific research toolkit:

- Python: `numpy`, `matplotlib`, `astropy`, etc.
- Machine learning: `tensorflow` (`keras`), `scikit-learn`, `jax`
- AI assistants: Antigravity, Claude, Gemini, Deep Research, CamberCloud
- Parallelization: OpenMP, GPU (CUDA C/C++, `pycuda`), cluster computing (`slurm/launcher`)
- General: `git`, L<sup>A</sup>T<sub>E</sub>X

## LANGUAGES

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**Russian:** Native.

**English/Portuguese:** Proficient.

**French:** Advanced.

**German:** Reading Knowledge.

## PRESENTATIONS

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| “Gravitational-wave source mixing in LISA” (talk), <i>WVU Center for Gravitational Waves and Cosmology Symposium</i> , Morgantown, WV                                                                           | 2026 |
| “Galactic Double White Dwarfs: A Case of Cosmic Confusion” (invited seminar), <i>PLASTRO REU program</i> , West Virginia University, Morgantown, WV                                                             | 2026 |
| “Confusion foregrounds of DWD populations in the UCB Catalogs project” (poster), <i>16th International LISA Symposium</i> , College Park, MD                                                                    | 2026 |
| “Hankel low-rank matrix approximation for time series denoising in gravitational-wave science” (talk), <i>APS Global Physics Summit 2026</i> , Denver, CO                                                       | 2026 |
| “Hankel low-rank matrix approximation for time series denoising” (talk), <i>NANOGrav Fall 2025 Meeting</i> , Missoula, MT                                                                                       | 2025 |
| “LISA double white dwarf binaries as Galactic accelerometers” (talk), <i>11th Gulf Coast Gravity Meeting</i> , University of Mississippi, Oxford, MS                                                            | 2025 |
| “LISA double white dwarf binaries as Galactic accelerometers” (contributed talk), <i>Ten Years to LISA Workshop</i> , NASA JPL, Pasadena, CA                                                                    | 2025 |
| “Inferring a population of intermediate-mass black holes with LISA” (talk), <i>245th Meeting of the American Astronomical Society</i> , National Harbor, MD                                                     | 2025 |
| “Quasimonochromatic LISA sources in the Frequency Domain” (talk), <i>APS April Meeting 2024</i> , Sacramento, CA                                                                                                | 2024 |
| “Inferring a Population of Intermediate-Mass Black Holes with LISA” (poster), <i>APS April Meeting 2024</i> , Sacramento, CA                                                                                    | 2024 |
| “Quasimonochromatic LISA sources in the frequency domain” (talk), <i>6th Neighborhood Workshop on Astrophysics and Cosmology</i> , Penn State University, State College, PA                                     | 2024 |
| “LISA Constraints on an Intermediate-Mass Black Hole in the Galactic Center” (talk), <i>MODEST-23: Star Clusters in the Post-Pandemic Era</i> , Evanston, IL                                                    | 2023 |
| “LISA Constraints on an Intermediate-Mass Black Hole in the Galactic Center” (talk), <i>APS April Meeting 2023</i> , Minneapolis, MN                                                                            | 2023 |
| “Hunting for intermediate-mass black holes with LISA binary radial velocity measurements” (talk), <i>Intermediate-Mass Black Holes: New Science from Stellar Evolution to Cosmology</i> , San Juan, Puerto Rico | 2022 |
| “Hunting for intermediate-mass black holes with LISA binary radial velocity measurements” (virtual talk), <i>23rd International Conference on General Relativity and Gravitation (GR23)</i> , Liyang, China     | 2022 |
| “Hunting for intermediate-mass black holes with LISA binary radial velocity measurements” (talk), <i>APS April Meeting 2022</i> , New York, NY                                                                  | 2022 |
| “The orbital Lense–Thirring precession in a strong field” (talk), <i>10th Alexander Friedmann International Seminar on Gravitation and Cosmology</i> , St. Petersburg, Russia                                   | 2019 |
| “Exoplanetary candidates to observations with Millimetron” (talk), <i>42nd COSPAR Scientific Assembly</i> , Pasadena, CA                                                                                        | 2018 |

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| “The full cycle of simulating black hole shadows: from ray tracing to spectrum-specific images of an arbitrarily shaped screen” (poster), <i>42nd COSPAR Scientific Assembly</i> , Pasadena, CA | 2018 |
| “Nonstandard objects and theories of gravity” (talk), <i>XXXIII Conference on Topical Problems of Extragalactic Astronomy</i> , Pushchino, Russia                                               | 2016 |
| “Numeric integration of geodesic equations in Schwarzschild’s field in different coordinate frames” (poster), <i>59th MIPT Scientific Conference</i> , Moscow, Russia                           | 2016 |
| “Black-and-white holes” (talk), <i>12th International Conference on Gravitation, Astrophysics and Cosmology (ICGAC-12)</i> , Moscow, Russia                                                     | 2015 |
| “Black-and-white hole as a space-time with integrable singularity” (talk), <i>9th Alexander Friedmann International Seminar on Gravitation and Cosmology</i> , Moscow, Russia                   | 2015 |
| “Phenomenological Approach in the Study of Singularity Problem” (contributed presentation), <i>26th Texas Symposium on Relativistic Astrophysics</i> , São Paulo, Brazil                        | 2012 |